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Radio LAB 3: Measuring the Milky Way's Rotation Curve

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1. INTRODUCTION

In this lab, we will measure the rotation curve of the Milky Way. We can measure galactic rotation with radio observations because radio waves are easy to detect everywhere in the Milky Way, even the far outskirts. Furthermore, radio waves are not affected by interstellar extinction, scattering of light, and the dust absorption leading to dark patches.

Using this galactic rotation curve, we can plot velocity as a function of galactic longitude to produce a $l - V_r$ diagram. This diagram will show objects with velocities differing from the galactic rotation (e.g., high velocity clouds). With this curve, we can also estimate the galactocentric distance to a particular star or cloud. Lastly, we can also estimate the mass of the Milky Way. This makes the galactic rotation curve a very useful piece of information.

We employ an important method called the tangent-point method. Since the method of trigonometric parallax to measure distances to astronomical objects is limited to relatively close objects, we use points in the Milky Way disk to estimate distance based on simple geometry. Tangent points only exist for certain longitudes and some have expected velocities close to 0 km/s while others are too close to the galactic center. Thus, using longitudes between 20 and 75 degrees, we can collect measurements by every full-width half maximum (6° for the small radio telescope) for the best results.

2. OBSERVATIONS

To collect data for this lab, we submitted an observing script (Figure 1) to the SRT on the Sterling Hall rooftop. We set the central frequency to 1420.4 MHz with observing mode 4. We covered a range of galactic positions from a longitude of 20° to 74°, increasing by 6° increments for a total of 10 positions. From the first position, we offset by 30° in elevation to run the temperature calibration. For longitudes under 40° we centered our spectra at 1420.3 MHz while for longitudes greater than 40° we centered our spectra at 1420.4 MHz. We ran a 300 second integration time for each longitude position. Observations were collected on November 11, 2025.

```
: record radio3_data_lauren.rad
: freq 1420.4 4
: galactic 20 0
: offset 0 30
: noisecal
: offset 0 0
: freq 1420.3 4
: galactic 20 0
:300
: galactic 26 0
:300
: galactic 32 0
:300
: galactic 38 0
:300
: freq 1420.4 4
: galactic 44 0
:300
: galactic 50 0
:300
: galactic 56 0
:300
: galactic 62 0
:300
: galactic 68 0
:300
: galactic 74 0
:300
: roff
```

Figure 1. SRT observing script

During observation, we manually recorded the VLSR value displayed on the SRT interface as this information is not included in the output data file. These values for each position are shown in Table 1. Throughout the 300 second integration time, these values changed by roughly ±0.1 km/s.

Galactic longitude	**VLSR**
20°	6.5 km/s
26°	6.0 km/s
32°	5.5 km/s
38°	4.9 km/s
44°	4.3 km/s
50°	3.6 km/s
56°	2.8 km/s
62°	2.1 km/s
68°	1.3 km/s
74°	0.5 km/s

Table 1. VLSR recordings

3. DATA ANALYSIS

Note: The Python code used for analysis is shown at the end in the Appendix (Section 6).

3.1. *Data Preparation*

Similarly to previous radio labs, I read the data file into Python while ignoring commented lines and labeling data columns as detailed in the UW SRT Manual. By comparing the timestamps in the data file with the read-in Python dataframe, I separated the data into calibration data (where the calibration diode was ran) and observing data (collected at each specified galactic longitude). The observing data will be used for analysis. Continuing with inspecting the timestamps, I separated the data collected at each galactic longitude beginning after the 300 second integration line. This will be used for later analysis.

3.2. *Time-Averaged Spectra*

For each of the 156 frequency channels, I took the average across all observations to get the average spectra values. Using the initial frequency ($freq_1 = 1419.71$ MHz) and the frequency channel width ($\Delta = 0.0078125$), I calculated a frequency array such that the ith frequency is given by

$$freq_i = freq_1 + i \cdot \Delta$$

resulting in 156 frequency values. We can visualize the average spectra by the frequency values (Figure 2).

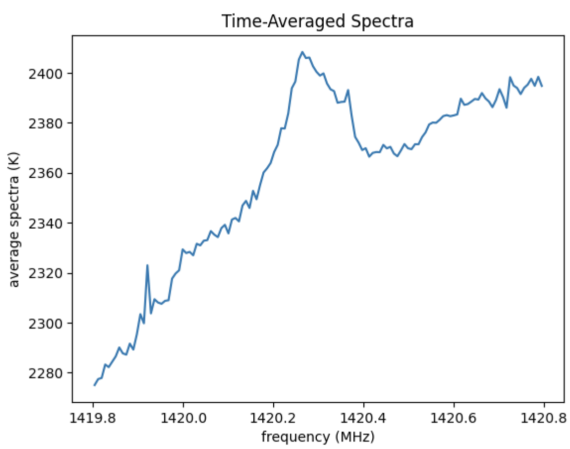

Figure 2. Average HI emission by frequency

We can also consider this relationship in terms of radial velocity. With the non-relativistic Doppler shift equation (1), we can convert frequency to radial velocity:

$$v = \frac{c(freq_0 - freq_{obs})}{freq_{obs}} \tag{1}$$

where c is the speed of light (in km/s), $freq_0$ is the rest frequency (1420.4 MHz), and $freq_{obs}$ corresponds to the values in the calculated frequency array.

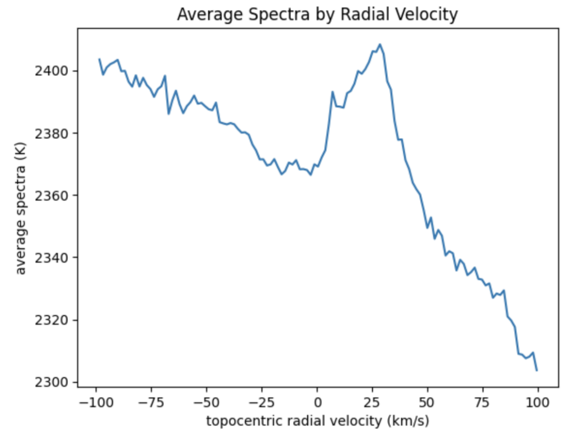

Figure 3. Average HI emission by radial velocity

3.3. *Linear Polynomial Fit*

Following from the previous lab, we performed a linear polynomial fit to spectral regions with no significant HI emission. By observing a plot of the full range of emission, I selected frequency channels with minimal emission to include in the fit. The baseline values were then calculated and subtracted from the original spectra. This change can be seen below in Figure 4

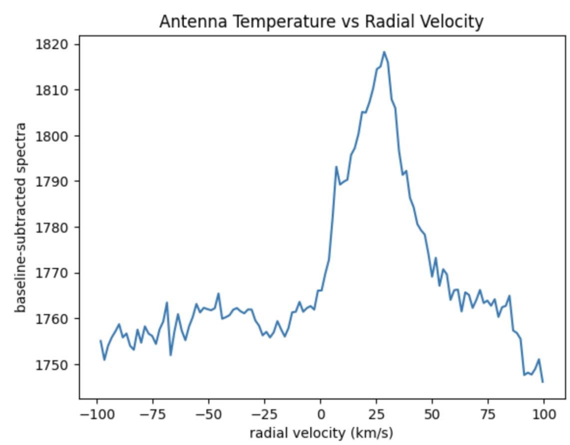

Figure 4. Baseline-subtracted spectra

3.4. *Tangent-Point Method*

In this subsection, we will employ the tangent-point method, broken down into further subsections for clarity.

3.4.1. *Velocity ($V_{r,T}$)*

For each galactic longitude, we can estimate the radial velocity corresponding to the tangent point along this line of sight. We will use two methods to estimate this velocity.

First, we consider the frequency channels with minimal HI emission, as we did above for the baseline fitting. From this, we calculate the noise in our spectrum (σ) and consider values above the threshold of 3σ. Then, the largest velocity corresponding to spectra above this threshold will be $V_{r,1}$.

Second, we can visualize the HI spectrum with data taken at each galactic longitude. The highest velocity corresponding to one of the clear peaks in the spectrum will be $V_{r,2}$. To achieve this, I considered the upper-edge cases and found the peaks with the highest spectra values for velocities between 75 and 100 km/s. The plots below show the average spectra for each longitude. Each has a dotted line at the clear peak with high velocity.

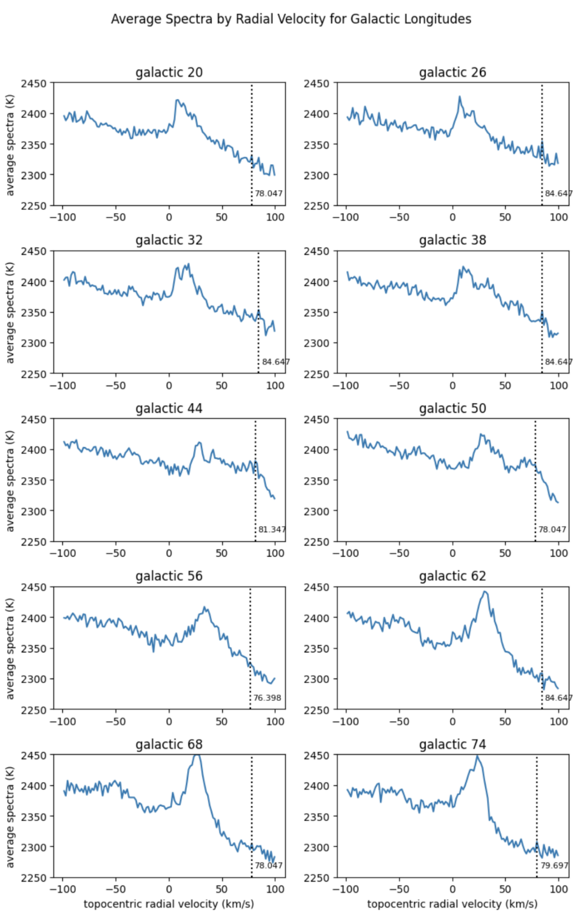

Figure 5. HI emission by galactic longitudes

Using these values, we can calculate the radial velocity corresponding to the tangent point:

$$V_r = \frac{V_{r,1} + V_{r,2}}{2}$$

and its uncertainty:

$$\sigma_{V_R} = \frac{V_{r,2} - V_{r,1}}{2}$$

. Then, we subtract 12.3 km/s from the calculated V_r to account for random motions of clouds in the line of sight to isolate contribution from galactic rotation.

3.4.2. *Distance (R)*

For each galactic longitude (ℓ), we can estimate the distance between the tangent point and the Galactic center using

$$R = R_\odot sin(\ell)$$

where $R_\odot = 8.5$ kpc. The final values for each distance/radius can be found in 8.

3.4.3. *Circular Velocity (V)*

Finally, we can calculate the circular velocity via

$$V = V_r + V_\odot sin(\ell)$$

where $V_\odot = 220$ km/s. Using error propagation, we can estimate the uncertainty, which simplifies to the previously calculated noise in V_r. The final values for each circular velocity can be found in 8.

3.5. *Galactic Rotation Curve*

Using the results for distances and velocities, we can plot the estimated galactic rotation curve. In Figure 6, each point corresponds to the galactic longitudes and the calculated values with uncertainties plotted. I also fit a third degree polynomial to model the curve.

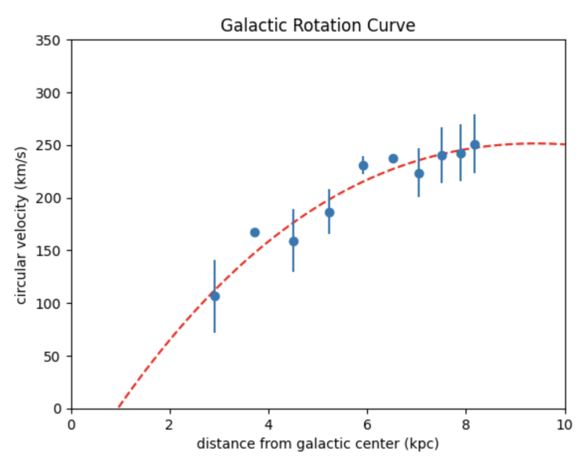

Figure 6. Galactic Rotation Curve: Our Result

The published results of Mróz et al. (2019) are shown in Figure 7. This plot combines data from 3 different studies, which gives us a good idea of others' results. While this plot has data with distances from 0 to 20 kpc, considering a range similar to ours (0 to 10 kpc) shows comparable results. Both plots start with points with velocities under 200 km/s which seem to increase and stabilize around 250 km/s.

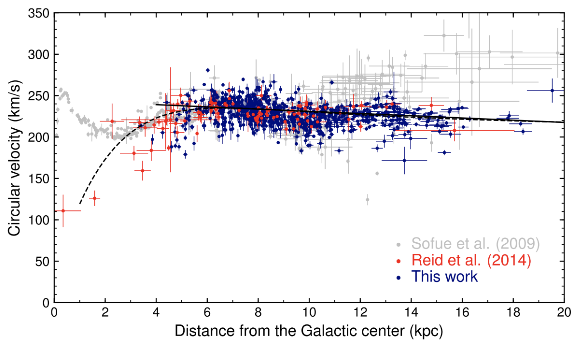

Figure 7. Galactic Rotation Curve: Published Results

3.6. *Mass as a Function of Galactocentric Radius*

With all of the above information, we can estimate the total mass of the Milky Way within a radius:

$$M_R = \frac{V^2 R}{G}$$

where G is the gravitational constant. After proper conversions to similar units, we can calculate the mass. The values for radius, velocity, and mass are displayed below for each galactic longitude where measurements were recorded.

longitude [deg]	radius [kpc]	velocity [km/s]	mass [kg]
20	2.907171	118.654339	1.892268e+40
26	3.726155	179.712963	5.563710e+40
32	4.504314	171.537325	6.127604e+40
38	5.233123	198.937972	9.575051e+40
44	5.904596	242.685969	1.607772e+41
50	6.511378	250.149457	1.883722e+41
56	7.046819	235.692674	1.809798e+41
62	7.505055	252.502029	2.212221e+41
68	7.881063	254.811306	2.365740e+41
74	8.170724	263.133326	2.615514e+41

Figure 8. Summary Table

4. CONCLUSIONS

I was able to achieve the main goal of this lab, by creating a galactic rotation curve for the galactic longitudes where I collected data. I feel that most of my analysis steps were accurate and similar to the previous lab. The comparison of my results with published results is promising, but mine could be improved. When selecting the highest velocity at tangent points, I was a bit unsure which peak to use. In other words, I was unsure of the correct balance between high radial velocity and spectra value. I tried a few different methods which resulted in slightly different galactic rotation curves, but these curves followed a similar pattern overall.

I tried to apply the velocity correction from topocentric to LSR, but got stuck. I recorded the VLSR values displayed by the SRT interface during data recording (Table 1), but was not sure how to incorporate these into my analysis. This likely would have resulted in an improved galactic rotation curve. The last improvement would have been to include data points from other classmates who collected measurements at different longitudes. One thing that could improve this experiment would be to assign longitude values to groups to ensure that we cover the full range from 20° to 75°.

This was a practical and useful application of data we collected. It emphasizes my amazement at turning numbers in a data file into meaningful contribtions to science!

5. REFERENCES

Mróz, P., Udalski, A., Skowron, D. M., Skowron, J., Soszyński, I., Pietrukowicz, P., Szymański, M. K., Poleski, R., Kozłowski, S., & Ulaczyk, K. (2019). Rotational curve of the Milky Way from classical cepheids. *The Astrophysical Journal Letters, 870*(1). https://doi.org/10.3847/2041-8213/aaf73f

6. APPENDIX

A. Python code is attached.